4.2 - Related Rates

Goals: Relate changing quantities by differentiating equations with respect to time.
Before Class Videos: 4.2.1

Example 1: An inverted conical tank has radius 6 ft and height 8 ft. Water is pumped in at 4 ft^3/min. How fast is the water level rising when the water is 4 ft deep?

Example 2: A rocket rises vertically at 600 m/s. An observer stands 4000 m from the launch point. How fast is the angle of elevation changing when the rocket is 3000 m high?

Method:

Example 3: Two boats are 400 km apart horizontally. Boat A goes north at 30km/h while boat B goes south at 45 km/h. How fast is the distance between them changing after 2 hours?

Example 4: A point moves along the curve $y = x^2$, with $x > 0$. Find the rate of change of its distance from $(3, 0)$ in terms of x and dx/dt.

Example 5: Suppose a bar of length d leans against a wall and slides along the ground. If $x'(t)$, the rate of change of its horizontal position is constant, is $y'(t)$ constant?

Example 6: Suppose a $2m$ tall skateboarder is a distance x from a $5m$ tall wall which has a light at the top. If $x'(t) = 2m/s$, how fast is the length of the shadow cast by the light increasing?

Example 7: A trough is 20 m long, with right triangular ends 5 m high and 5 m wide. Water flows in at 100 m^3/h. How fast is the water level rising when the water is 2 m deep?

Suggested after class problems from APEX: Section 4.2 problems 3,5,7